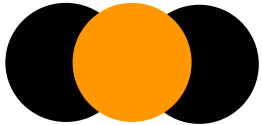


Syracuse Housing Code Violations Dashboard



Project Overview

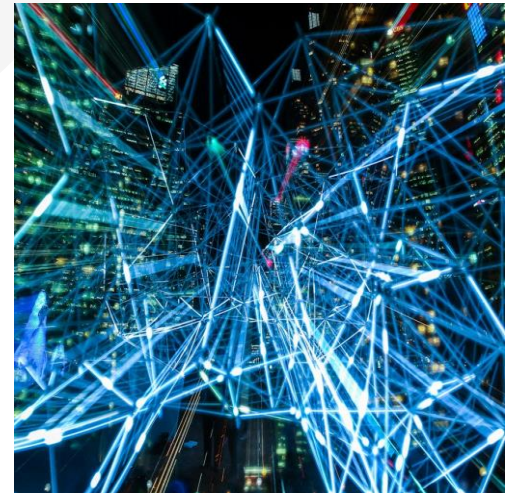
01 Housing conditions impact safety and quality of life

02 Data exists but is difficult to interpret interpret

03 Focus on violations, patterns, and and system performance

04 Stakeholder-focused civic insights insights

05 Dashboard for real-world usability





Problem & Research Question

The Problem

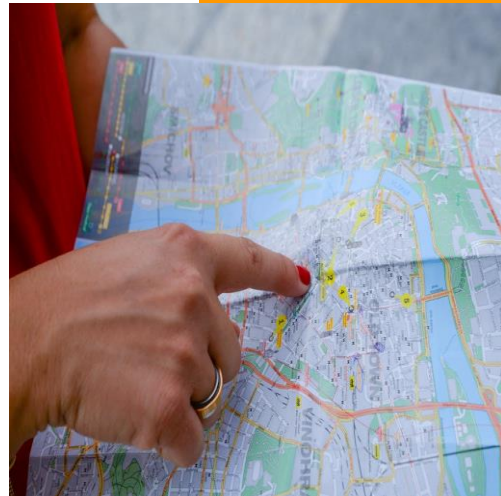
Housing conditions directly affect safety, neighborhood stability, and quality of life. While Syracuse provides open data on code violations, the raw dataset is difficult to interpret without structured analysis and visualization.

Research Question

This project asks how violations are distributed across neighborhoods and time, and what and what patterns in violation type, case status, and resolution time reveal about system performance.

Why It Matters

Clear interpretation of this data enables stakeholders to understand community needs, needs, evaluate performance, and make informed decisions that improve housing conditions.





Data & Challenges

Dataset Overview

The analysis uses the Syracuse Open Data Code Violations dataset, containing over 143,000 records with details on violation type, status, dates, and neighborhood location.

Data Quality Issues

Challenges include missing values, inconsistent formatting, and mixed date fields. Additionally, violation reporting is complaint-driven, which introduces bias in how issues are captured.

Implications

These limitations require careful interpretation. The analysis focuses on descriptive patterns rather than causal conclusions, ensuring responsible use of civic data.





Approach & Methodology

Analysis Pipeline

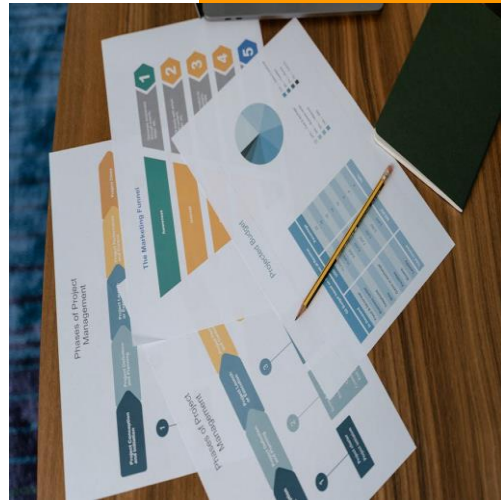
The project follows a structured pipeline from raw data to cleaned datasets, analysis, analysis, validation, and final dashboard output. Each step is modular to ensure ensure reproducibility and clarity.

Techniques Used

Data cleaning, feature engineering, and aggregation were performed using Python and Python and Pandas. Key features include year, case status, and resolution time for time for analysis.

LLM Validation

LLMs were used only for documentation and narrative support. All insights and findings and findings were validated using computed statistics to ensure accuracy and avoid and avoid hallucinations.



Key Findings Overview

01 Violations concentrated in specific neighborhoods

02 Few violation types dominate

03 Trends increase over time

04 Most cases resolved but backlog backlog exists

05 Resolution time varies across neighborhoods





Neighborhood Patterns

Observation

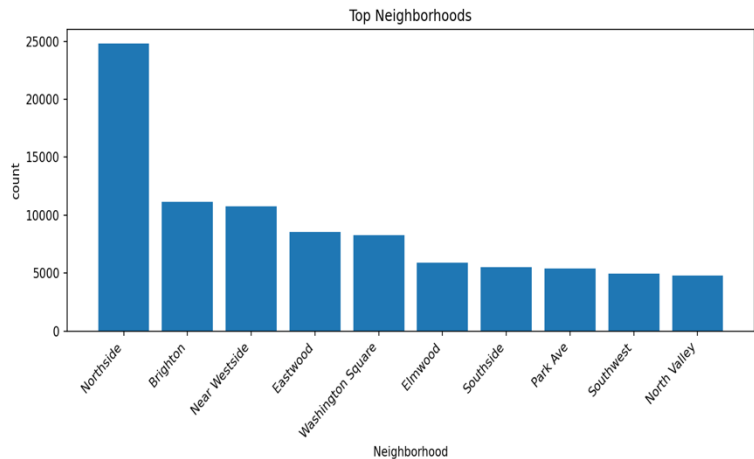
Violations are heavily concentrated in specific neighborhoods such as Northside, Brighton, and Near Westside. This uneven distribution highlights areas with higher housing challenges.

Interpretation

The concentration may reflect differences in housing conditions, enforcement, or reporting behavior. These patterns suggest the need for need for targeted attention in specific areas.

Impact

City officials can use this insight to allocate resources, prioritize inspections, and design interventions for neighborhoods with the highest needs.





Violation Types & Trends

Key Patterns

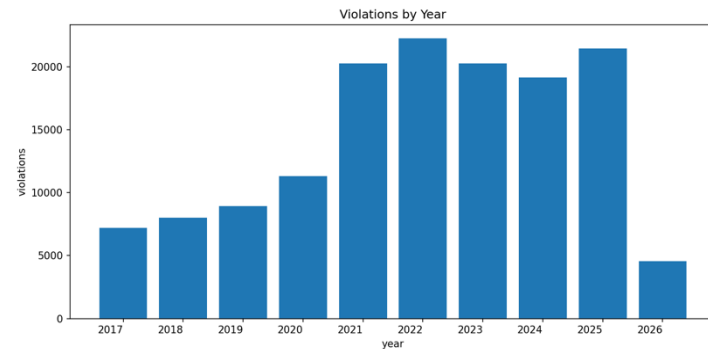
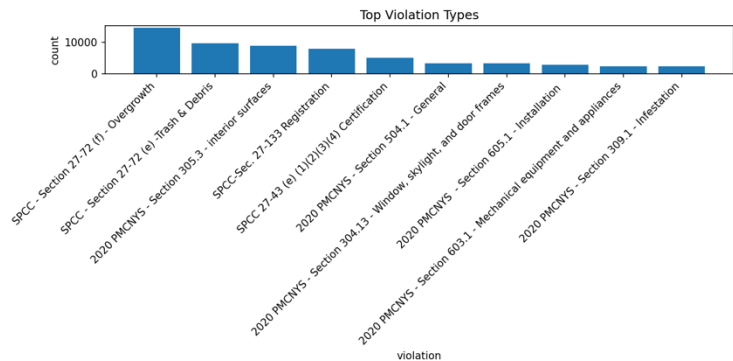
A small number of violation types, particularly overgrowth and trash- and trash-related issues, account for a large share of all reported cases. This indicates recurring problems rather than isolated incidents.

Trends Over Time

The number of violations has increased over time, peaking around recent years. This may reflect changes in reporting, enforcement, or actual housing conditions.

Impact

Addressing the most common violation types could significantly reduce total reported cases and improve overall neighborhood conditions.

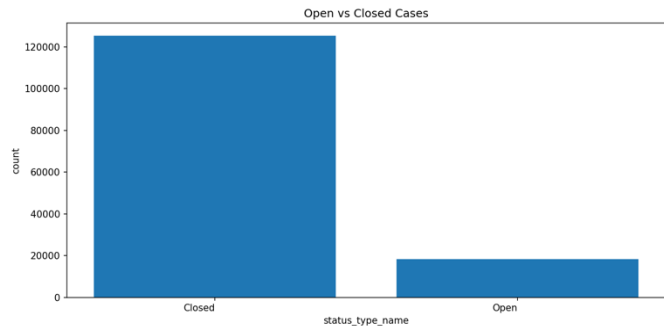




System Performance

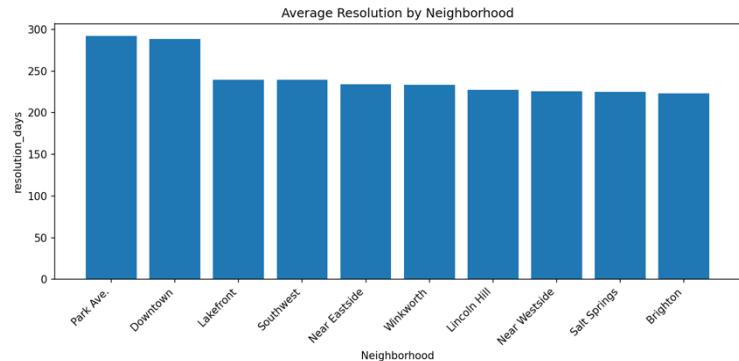
Case Status

Most violations are eventually resolved, with a majority of cases marked as closed. However, a notable number of open cases remain, indicating ongoing workload.



Resolution Time

Resolution time varies across neighborhoods, suggesting differences in service efficiency or case complexity. Some areas experience longer delays than others.



Impact

Improving resolution time and reducing open cases can enhance public trust and ensure more equitable service delivery across the city.



Limitations & Responsible Interpretation

Key Limitations

The dataset reflects reported violations rather than all housing issues. Complaint-Complaint-driven reporting and inconsistent data quality can introduce bias and affect bias and affect comparisons across neighborhoods.

Interpretation Approach

This analysis focuses on descriptive patterns rather than causal relationships. Findings relationships. Findings are presented with caution to avoid overgeneralization or overgeneralization or misleading conclusions.

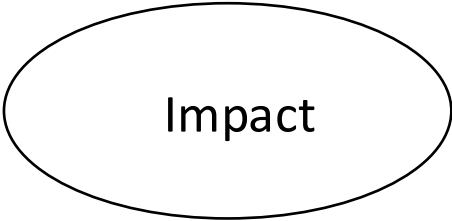
Future Work

Future improvements could include integrating additional datasets, improving data improving data quality, and developing predictive models to support proactive proactive decision-making.

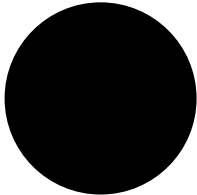




Conclusion & Civic Civic Impact



Impact



What value does this project
provide?

This project transforms Syracuse open data into clear, actionable actionable insights that support residents, policymakers, and and community organizations. By improving transparency and and highlighting patterns in housing conditions, it contributes to contributes to better-informed decision-making and more effective city services.

